

SEMESTER S5

OPERATIONS RESEARCH

Course Code	PBCBT504	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with knowledge and skills needed to apply various operations research tools and techniques for decision making in organizations.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	INTRODUCTION TO OR & LINEAR PROGRAMMING Origin of OR and its definition – Types of OR problems – Deterministic vs stochastic optimization, Phases of OR – Problem formulation – Building mathematical model – Deriving solutions, Validating model, Controlling and implementing solution. Linear programming – Example from industrial cases, Formulation of LPP – Implicit assumptions of LPP. Solution of LPP – Graphical method, Simplex method – Artificial variables – Big M method – Identification & resolution of special cases – Infeasibility, Unboundedness, Redundancy & Degeneracy – Sensitivity analysis.	11
2	TRANSPORTATION & ASSIGNMENT PROBLEMS Transportation problem – Introduction – Mathematical formulation – Balanced & Unbalanced	9

	Transportation Problems – Initial basic feasible solution – North-west corner rule – Lest cost method – Vogel’s approximation method – Test for optimality – MODI method – Degeneracy & its solution. Assignment problem – Introduction – Mathematical formulation – Balanced & Unbalanced Assignment Problem. Solution method – Hungarian algorithm	
3	INVENTORY CONTROL Functions of inventory and its disadvantages, ABC analysis, Concept of inventory costs, Basics of inventory policy – Order, lead time. Fixed Order Quantity, Economic Order Quantity (EOQ), Production Order Quantity (POQ).EOQ models (manufacturing and purchase) with shortages and without shortages.	11
4	QUEUING THEORY Definitions – Queue (waiting line), waiting time costs, characteristics of queuing system (arrival, service & queue discipline), queue types (channel vs phase). Kendall’s notation, Little formula, steady state behavior, Poisson’s process & queue, Queuing models – and its performance measures with examples – Brief description about some special models (M/M/1/N/FCFS, M/M/1/∞/FCFS). Project definition, project scheduling techniques – PERT & CPM, Grant chart – Determination of critical paths, Estimation of project time and its variance in PERT. Concept of project crashing / time – Cost trade off. Simulation methodology: Definition and steps of simulation, random number, random number generator, Application in scheduling, Queuing systems and inventory systems.	13

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	2 questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 2 sub divisions. - Each question carries 6 marks. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Formulate and solve Linear Programming problems. .	K3
CO2	Model and solve transportation and assignment problems.	K3
CO3	Apply various aspects of inventory management and compute optimum inventory order quantities and safety levels.	K3
CO4	Derive the mathematical models of queues and compute various measures of performance in these models	K3
CO5	Construct networks and find optimal scheduling using CPM and PERT.	K3
CO6	Apply simulation techniques in Project scheduling, Queueing and Inventory systems.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3	3	3					3	3
CO2	3	3	3	3	3	3					3	3
CO3	3	3	3	3	3	3					3	3
CO4	3	3	3	3	3	3					3	3
CO5	3	3	3	3	3	3					3	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Operation Research: An introduction	Hamdy A. Taha	Pearson	10 th Edition 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Linear Programming	Katta G. Murty	Wiley	1983
2	Linear Programming	G. Hadley	Addison-Wesley publishing company	1978.
3	Principles of Operation Research with Applications to Managerial Decisions	H.M Wagnes, Englewood Cliffs	Prentice Hall	1975
4	Introduction to Operation Research	F.S. Hillier and G. J. Lieberman	McGraw Hill	11 th edition, 2021
5	Elements of Queuing Theory, with Applications.	Thomas L. Saaty	McGraw – Hill, New York	1961
6	Operation Research (Fourth Edition)	S Kalavathy	Vikas Publishing House Pvt.Ltd	2013
7	Operations Research and Management Science, Hand Book	A.Ravi Ravindran	CRC press	1st edition, 2007.
8	Management Guide to PERT / CPM	J.D. Wiest and F.K. Levy	Prentice –Hall, NJ	1969

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches